

L3 – Transformations of Sine and Cosine Part 1

Equation → Graph

MCR3U

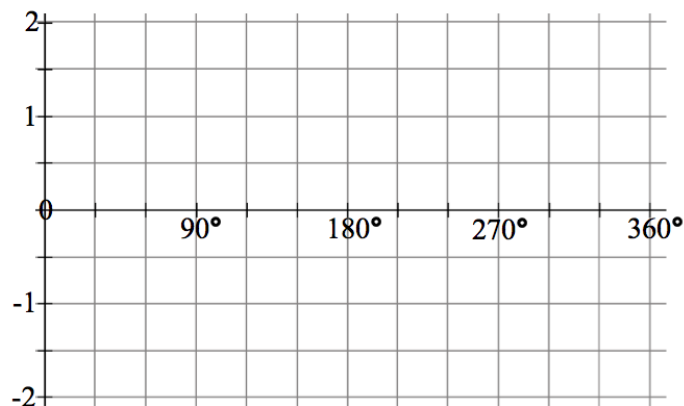
Section 1: Review of Sine and Cosine Functions

$$y = a \sin[k(x - d)] + c \text{ OR } y = a \cos[k(x - d)] + c$$

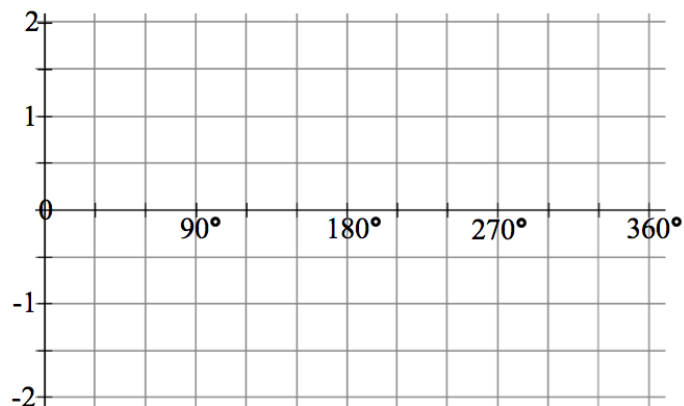
a	k	d	c
Vertical stretch or compression by a factor of a . Vertical reflection if $a < 0$ $ a = \text{amplitude}$	Horizontal stretch or compression by a factor of $\frac{1}{k}$. Horizontal reflection if $k < 0$. $\frac{360}{ k } = \text{period}$	Phase shift $d > 0$; shift right $d < 0$; shift left	Vertical shift $c > 0$; shift up $c < 0$; shift down

Graphs of parent functions $y = \sin x$ and $y = \cos x$ using key points:

x	y

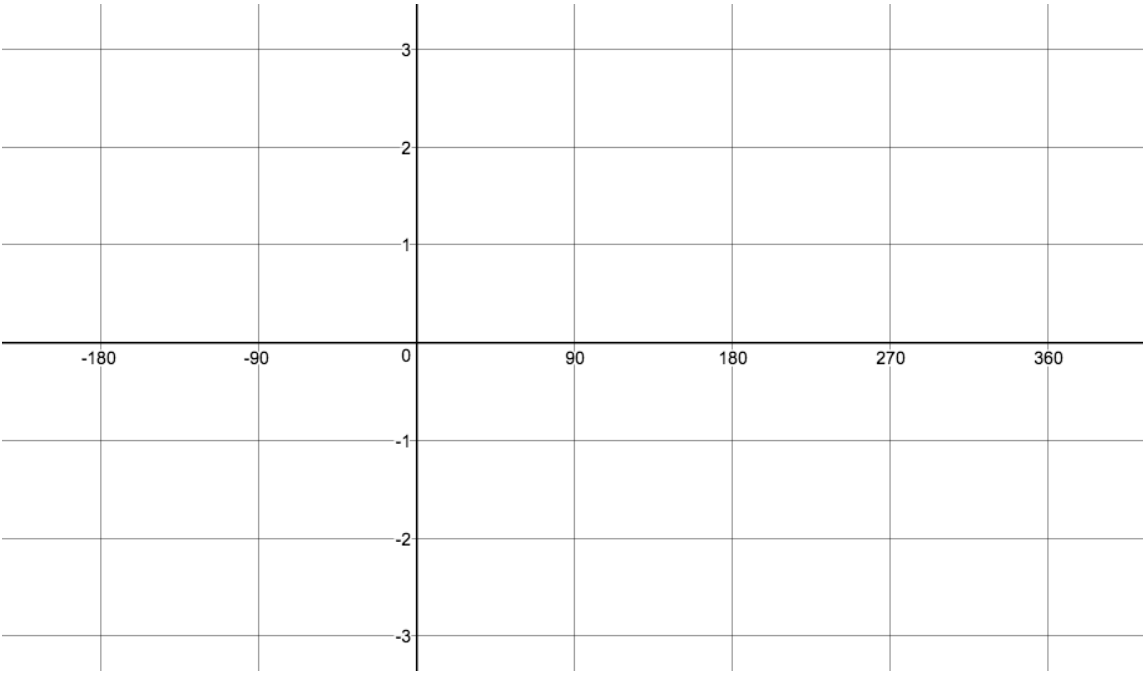


x	y



Section 2: Graphing Transformed Sinusoidal Functions

Example 1: Graph $y = 2 \sin x + 1$ using transformations. Then state the amplitude, period, and number of cycles between 0° and 360° .

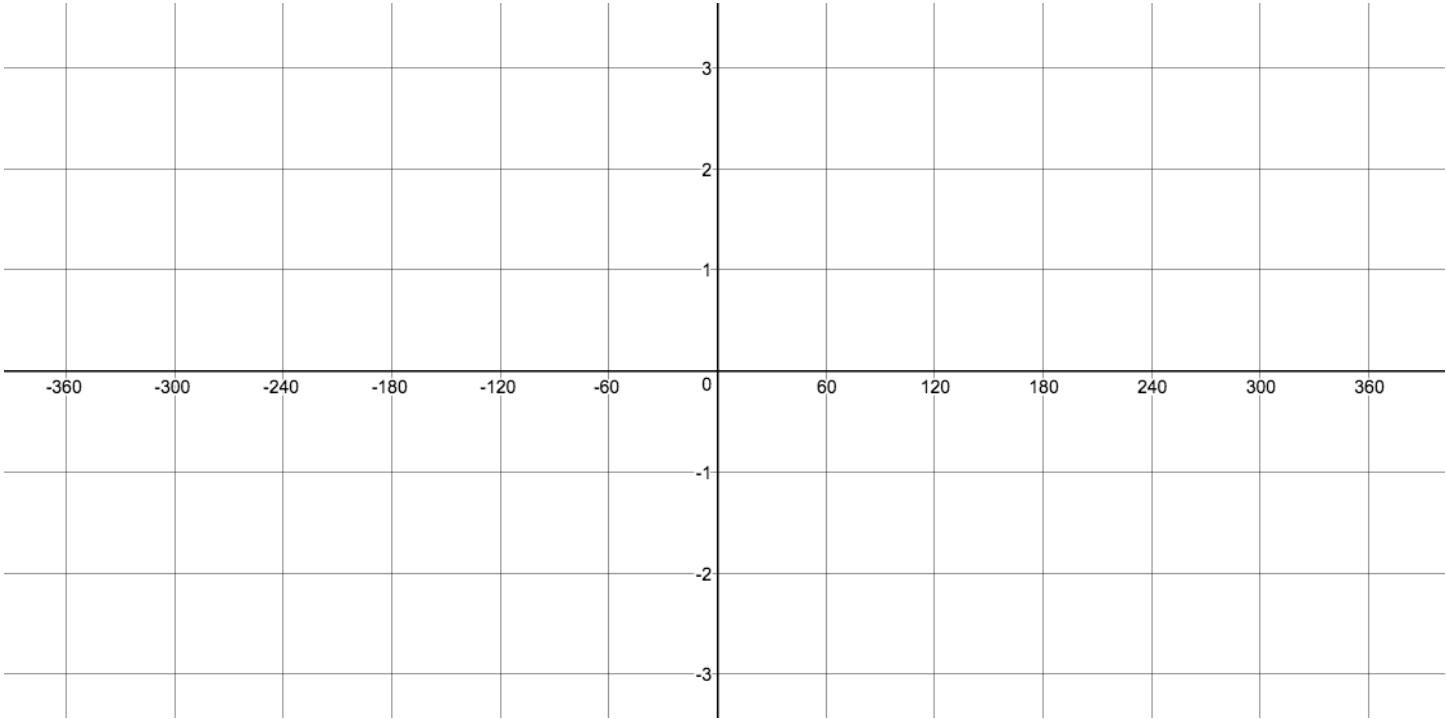


Amplitude:

Period:

Number of cycles between 0° and 360° :

Example 2: Graph $y = -1.5 \cos[3(x - 30^\circ)] + 0.5$ using transformations. Then state the amplitude, period, and number of cycles between 0° and 360° .

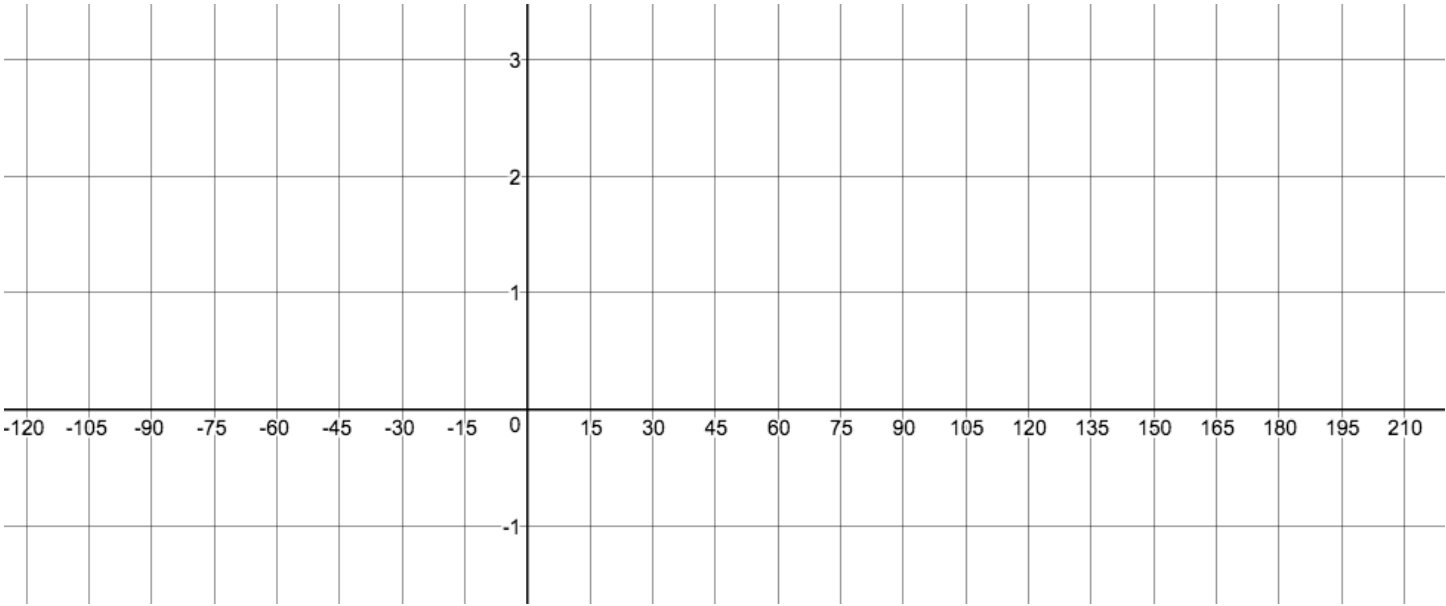


Amplitude:

Period:

Number of cycles between 0° and 360° :

Example 3: Graph $y = \sin[-4(x - 60^\circ)] + 2$ using transformations. Then state the amplitude, period, and number of cycles between 0° and 360° .



Amplitude:

Period:

Number of cycles between 0° and 360° :